

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 1 – Constraint-Based Problem Solving
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
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Section / Batch:	B.Tech AI &DS	Date:	14-08-2026

Code of Conduct

I _____ N.Kishan Kumar _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____ N.Kishan Kumar _____

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse, Reference Resolution & Text Coherence	Apply constraint-based problem solving for reference resolution, identify agreement and coherence constraints, resolve coreferences, and ensure entity chain consistency in a given paragraph.	Q1
Dialog and Conversational Agents, Discourse Planning & Surface Realizations	Construct constrained dialog responses by applying dialog acts, discourse planning, coherence and politeness constraints, generate multiple valid responses, and evaluate the best output.	Q2
Machine Translation, Discourse Structure, Predicate Logic & Interpretation	Perform word sense disambiguation using constraints, construct predicate logic representations, maintain discourse structure (rhetorical relations), and generate coherent translations/paraphrases.	Q3

Annexure A – Constraint-Based Problem Solving

1. Constraint-Based Coreference Resolution

Given Paragraph

“John and Mary went to the park. He brought a ball. She wanted to play with it. The dog chased him excitedly. Finally, they all went home.”

a) Identification of Referring Expressions and Possible Antecedents

A **referring expression** or **mention** is a word or phrase that refers to an entity introduced in the discourse. In the given paragraph, several pronouns such as *he*, *she*, *it*, *him*, and *they* need to be resolved to their appropriate antecedents.

Referring Expression	Possible Antecedents	Final Antecedent
He	John, Mary	John
She	John, Mary	Mary
it	ball, park, dog	ball
him	John, Mary, dog	John
they	John, Mary, dog	John, Mary, dog

The proper nouns **John**, **Mary**, and **dog** are explicitly introduced entities. The pronouns refer back to these entities.

b) Applying Constraints for Reference Resolution

1. Gender and Number Agreement

Gender and number agreement eliminate several incorrect possibilities.

- He is singular and masculine, so it refers to John, not Mary.
- She is singular and feminine, so it refers to Mary, not John.
- It is singular and normally refers to a non-human object. Therefore, it refers to the ball rather than John, Mary, or the dog.
- Him is singular and masculine, so John is a possible antecedent.
- They is plural, so it cannot refer to only John, Mary, or the dog individually. It can refer to the group of people or all entities mentioned.

2. Recency Constraint

The recency constraint prefers an antecedent that was mentioned recently.

For example, when resolving **it** in “She wanted to play with it,” the nearest suitable noun is **ball**, which was introduced in the previous sentence. Therefore, *it* → *ball*.

However, recency should not be applied blindly. A recent entity must also satisfy gender, number, semantic compatibility, and discourse coherence.

3. Semantic Compatibility

The meaning of the sentence also helps resolve references.

- Mary can want to play with a ball, so *it* → *ball* is semantically appropriate.
- A dog can chase a person, so the sentence “The dog chased him excitedly” strongly supports *him* → *John*.
- A dog cannot normally be referred to as him unless the dog is explicitly established as male. Therefore, without such information, John is the stronger interpretation.

4. Coherence

The final interpretations should produce a coherent sequence of events:

John and Mary → John → Mary → ball → John → John, Mary, dog

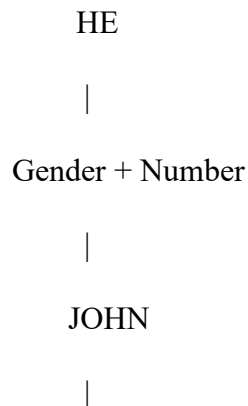
This creates consistent entity chains and makes the paragraph logically understandable.

c) Constraint Graph / Table

The following table shows how the constraints eliminate possible links.

Mention	Candidate	Gender/Number	Recency	Semantic Compatibility	Decision
He	John	✓	✓	✓	Selected
He	Mary	✗	✓	✓	Eliminated
She	Mary	✓	✓	✓	Selected
She	John	✗	✓	✓	Eliminated
It	Ball	✓	✓	✓	Selected
It	Park	✓	✗	Possible but weak	Eliminated
It	Dog	✓	✓	Weak	Eliminated
Him	John	✓	✓	✓	Selected
Him	Mary	✗	✓	✗	Eliminated
Him	Dog	Uncertain	✓	Possible but less suitable	Eliminated
They	John + Mary + Dog	✓	✓	✓	Selected
They	John only	✗	✓	✗	Eliminated
They	Mary only	✗	✓	✗	Eliminated

Simple Constraint Graph



Semantic / Recency

|

SHE

|

Gender + Number

|

MARY

IT

|

Semantic Compatibility

|

BALL

HIM

|

Gender + Semantics

|

JOHN

THEY

|

Plural Agreement

|

JOHN + MARY + DOG

d) Final Coreference Chains

The final coreference chains are:

- **John** → **He** → **him** → **they**
- **Mary** → **She** → **they**
- **ball** → **it**
- **dog** → **they**

Thus, the references can be represented as:

John = He = him

Mary = She

ball = it

John + Mary + dog = they

Rewritten Paragraph with Clear References

John and Mary went to the park. John brought a ball. Mary wanted to play with the ball. The dog chased John excitedly. Finally, John, Mary, and the dog all went home.

The rewritten paragraph removes ambiguity while preserving the original meaning.

e) Priority Order of Constraints

A suitable priority order is:

1. **Gender and Number Agreement**
2. **Semantic Compatibility**
3. **Coherence**
4. **Recency**

Gender and number agreement should receive high priority because they provide strong grammatical restrictions. For example, “she” cannot normally refer to John.

Semantic compatibility is also important because the selected antecedent must make sense in the context. A dog chasing a person is more natural than a dog chasing an unrelated object.

Coherence ensures that the references form a consistent story and entity chain.

Recency is useful as a tie-breaking constraint because the nearest suitable antecedent is often the correct one. However, it should not override stronger grammatical or semantic constraints.

Effect of Relaxing the Recency Constraint

If the recency constraint is relaxed, the system can still resolve references using gender, number, semantics, and coherence. However, when multiple candidates are equally compatible, the system may have difficulty selecting the intended antecedent.

For example, “it” could potentially refer to several previously mentioned objects or entities. Without recency, the system has fewer reasons to prefer the recently mentioned ball. Therefore, relaxing recency can increase ambiguity and computational search, although it may sometimes prevent an incorrect choice when the intended antecedent is not the most recent entity.

2. Constraint-Based Dialog Planning

Conversation History

User: “I have an important exam tomorrow but I’m not able to concentrate.”

Required Dialog Act

Advise + Encourage

The response should provide useful advice while motivating the student.

a) Relevant Discourse Planning Steps

Before generating the response, the system should perform the following steps:

Step 1: Identify the Main Entities

The important entities and concepts are:

- **you** – the user/student
- **exam** – the important upcoming event
- **concentrate** – the current difficulty

Step 2: Identify the User's Intention

The user is expressing difficulty in concentrating before an exam. Therefore, the response should provide **advice and encouragement**.

Step 3: Establish a Discourse Relation

A **Cause-Effect** relation can be used:

Difficulty concentrating

↓

Take a short break / reduce distractions

↓

Better focus

↓

Improved exam preparation

Step 4: Select Appropriate Keywords

At least two required keywords should be included:

- **focus**
- **break**
- **confident**

Step 5: Generate a Short Response

The response must contain only **2–3 sentences**, maintain a positive tone, and avoid contradictions.

b) Three Possible Responses

Response 1

Take a short break, remove distractions, and then study in small focused sessions so you can concentrate better for the exam. You have prepared for this, so stay confident and focus on one topic at a time.

Response 2

Since the exam is tomorrow, take a short break if you feel distracted and then return to your revision with a clear focus. Stay confident because even a few productive study sessions can make your exam preparation better.

Response 3

You may be finding it difficult to concentrate because the exam is important, so take a short break and come back with a fresh mind. Focus on the most important topics first, and remind yourself that you can handle the exam confidently.

c) Evaluation and Selection

Constraint	Response 1	Response 2	Response 3
2–3 sentences	✓	✓	✓
Maintains “exam”	✓	✓	✓
Maintains “concentrate”/focus idea	✓	✓	✓
Uses required keywords	✓	✓	✓
Advice	✓	✓	✓
Encouragement	✓	✓	✓
Logical consistency	✓	✓	✓
Positive tone	✓	✓	✓
Cause-Effect/Elaboration	✓	✓	✓

Best Response

Response 1 is selected as the best response.

It directly addresses the concentration problem, gives a practical solution through short study sessions and reduced distractions, and provides encouragement using “stay confident.” It also maintains entity coherence by referring to the exam and the student's ability to concentrate and focus.

d) Effect of Violating Two Constraints

Violation 1: Response Length Constraint

If the response exceeds 2–3 sentences, it may become unnecessarily long. Since the user is already having difficulty concentrating, excessive information may reduce clarity and make the advice less effective.

Violation 2: Keyword Constraint

If the response does not contain at least two of the required keywords such as **focus**, **break**, or **confident**, it would fail to satisfy the specified generation constraint. The response might still be meaningful, but it would not fully meet the requirements of the dialog-planning system.

Therefore, constraint satisfaction improves the relevance, coherence, consistency, and usefulness of generated dialogue.

3. Constraint-Based Word Sense Disambiguation

Source Sentence

“The bank by the river flooded after the storm, but it was saved by quick action.”

The word **“bank”** is ambiguous because it can refer to:

1. A **financial institution**, or
2. The **side of a river**.

a) Resolving the Ambiguous Word “Bank”

The correct meaning of **bank** is **riverbank**.

Justification

The phrase **“by the river”** provides strong contextual evidence that the word refers to the land beside the river.

The predicate **“flooded”** also supports the riverbank interpretation. A riverbank can naturally flood after a storm, whereas a financial bank does not normally flood simply because it is a bank.

Therefore:

bank = riverbank

The constraints support this interpretation:

Constraint	Evidence	Result
Context	“by the river”	Riverbank
Semantic compatibility	A riverbank can flood	Riverbank
Predicate compatibility	“flooded” suits riverbank	Riverbank
Coherence	Storm and flooding are related	Riverbank
Financial interpretation	Weak semantic connection	Eliminated

Thus, the financial institution sense of **bank** is eliminated.

b) Predicate Logic Representation

Let:

- **b** = the bank/riverbank
- **r** = the river
- **s** = the storm
- **a** = quick action

Predicates:

- **bank(b)** – b is a bank
- **river(r)** – r is a river
- **location(b,r)** – the bank is located beside the river
- **storm(s)** – s is a storm
- **flood(b)** – the bank flooded
- **after(b,s)** – the flooding occurred after the storm
- **action(a)** – a is an action
- **quick(a)** – the action was quick
- **save(a,b)** – the quick action saved the bank

A simplified predicate logic representation is:

$\text{bank}(b) \wedge \text{river}(r) \wedge \text{location}(b,r)$

$\wedge \text{storm}(s) \wedge \text{flood}(b)$

$\wedge \text{after}(\text{flood}(b), s)$

$\wedge \text{action}(a) \wedge \text{quick}(a)$

$\wedge \text{save}(a,b)$

The representation captures the important entities and relationships in the sentence.

c) Target Language / Simple English Paraphrase

Simple English Paraphrase

The riverbank near the river flooded after the storm, but quick action saved it from further damage.

This paraphrase:

- Resolves **bank** as **riverbank**.
- Preserves the storm and flooding information.
- Maintains the contrast between flooding and being saved.
- Preserves the relationship between the riverbank and the river.
- Improves clarity by explicitly identifying the meaning of “bank.”

Indian Language Example – Telugu

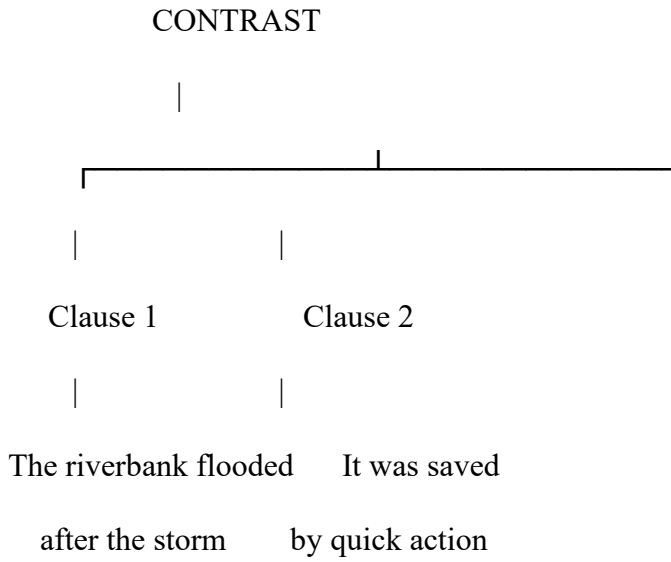
“తుఫాను తర్వాత నది పక్కనున్న ఒడ్డు వరదకు గురైంది, కానీ త్వరిత చర్యల వల్ల అది రక్షించబడింది.”

English meaning:

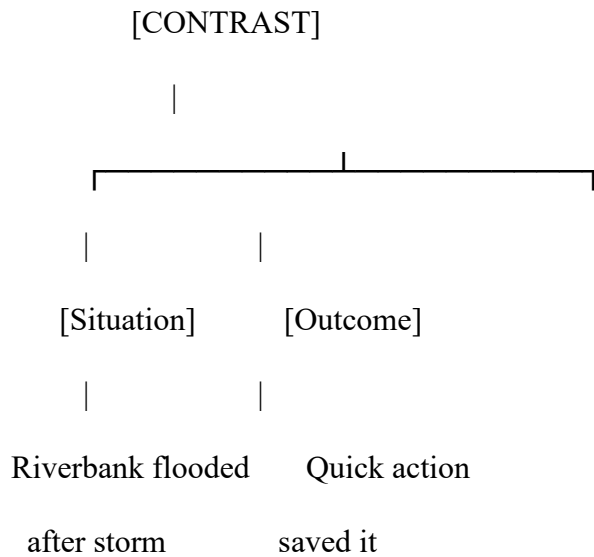
“After the storm, the riverbank was flooded, but it was saved through quick action.”

d) Simple RST-Style Discourse Tree

The sentence contains two clauses connected by “**but**.” Therefore, the main rhetorical relation is **Contrast**.



A more detailed representation is:



The first clause describes an adverse event, while the second clause describes a positive outcome. The word “but” signals the contrast between these two situations.

e) Advantages of Constraint-Based Approach Over Pure Statistical Machine Translation

A constraint-based approach explicitly uses linguistic, semantic, and contextual constraints to select the correct interpretation. This is particularly useful for ambiguous words such as “bank.”

1. Better Word Sense Disambiguation

The approach uses context such as “by the river” and “flooded” to identify the riverbank meaning rather than relying only on word frequency.

2. Preserves Important Information

Constraints ensure that important entities such as the bank, river, storm, and quick action are not accidentally removed or changed during generation.

3. Maintains Logical Consistency

The selected meaning must be compatible with the predicate “flooded.” This prevents an unlikely interpretation such as a financial institution being selected without supporting context.

4. Preserves Discourse Structure

The relationship between the two clauses is explicitly maintained as a Contrast relation. The meaning of “but” is therefore preserved.

5. More Explainable Decisions

Constraint-based systems can explain why a particular interpretation was selected. For example:

“by the river” + “flooded”

↓

Riverbank

A pure statistical approach may produce a likely translation based on learned patterns, but a constraint-based approach can explicitly show the linguistic reasons behind the decision.

Comparison

Aspect	Constraint-Based Approach	Pure Statistical MT
Word sense	Uses explicit context and rules	Uses learned statistical patterns
Ambiguity handling	Strong when constraints are well designed	May select frequent but incorrect sense
Logical consistency	Explicitly enforced	Not always guaranteed
Information preservation	Can be explicitly constrained	May occasionally omit or alter information
Discourse structure	Can be explicitly preserved	May not always preserve rhetorical relations

Explainability	High	Relatively low
Flexibility	Depends on manually defined constraints	Learns patterns from large data
Best suited for	Ambiguous and constraint-sensitive sentences	Large-scale general translation

Conclusion

Constraint-based problem solving is effective when a sentence contains ambiguity, strict grammatical requirements, semantic restrictions, or important discourse relationships. In the given example, the constraints correctly identify “bank” as riverbank, preserve the flooding and rescue events, and maintain the Contrast relation. Although statistical machine translation is powerful for large-scale translation, combining statistical methods with explicit constraints can provide more reliable and interpretable results for linguistically complex sentences.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Q. No. / Criteria Level	Problem Understanding	Constraint Analysis	Solution Development	Application of Concepts	Justification	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____